

Locally Robust Commensurate BART for Borrowing External Controls

Local compatibility and borrowing calibration

Yuan Ji

Department of Public Health Sciences
University of Chicago

Collaborators: Yunxuan Zhang (UChicago), Gu Mi (Sanofi), Ji Lin (Sanofi)
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Why borrow external controls?



What external controls can contribute

- ▶ More observations for the control response surface
- ▶ Greater precision when the external and trial populations agree
- ▶ A smaller concurrent control arm in settings where borrowing is justified

The gain depends on transportability.



Compatibility can vary across patients

An external control may agree with the trial for one patient profile and differ for another.

- ▶ Eligibility criteria and care pathways can differ by patient group.
- ▶ Measured and unmeasured confounding can create regional shifts.

One study-level borrowing parameter cannot distinguish those regions.

Established priors borrow at the study level

Prior	Borrowing mechanism
MAP prior	Exchangeable study parameters, with a robust component that reduces borrowing under conflict (Neuenschwander et al., 2010; Schmidli et al., 2014)
Commensurate prior	$\theta \mid \theta_0 \sim N(\theta_0, 1/\tau)$, with a spike-and-slab prior on τ (Hobbs et al., 2011, 2012)
Power prior	Historical likelihood raised to a discount parameter a_0 (Ibrahim and Chen, 2000)

LRC-BART lets borrowing change with the covariates.

With one historical study, these forms are closely related under suitable parameterizations (Röver and Friede, 2025).

A local discrepancy model



The trial-control mean is $f(\mathbf{x}) + g(\mathbf{x})$

RCT controls have $D_i = 1$ and external controls have $D_i = 2$:

$$y_i = f(\mathbf{x}_i) + \mathbb{I}\{D_i = 1\} g(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim \text{N}(0, \sigma_{D_i}^2).$$

$f(\mathbf{x})$ External-control response mean, fitted to both control sources

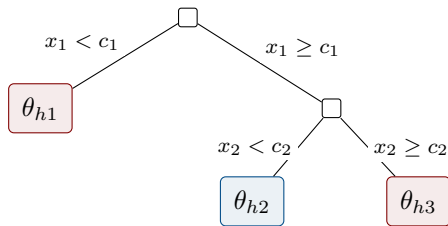
$g(\mathbf{x})$ Local discrepancy, informed by the RCT controls

The decomposition follows Bayesian causal forests, with source membership replacing treatment (Hahn et al., 2020).

The discrepancy changes across patient profiles

RWD controls: $f(\mathbf{x})$ **RCT controls:** $f(\mathbf{x}) + g(\mathbf{x})$

One tree in g (illustrative splits)



Each profile reaches one leaf per tree.
Add those contributions: $g(\mathbf{x}) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(\mathbf{x})}$.

f : the external response mean
Fitted jointly to both control sources.

Spike: $z_{h\ell} = 1$
 $\theta_{h\ell} \sim N(0, \tau_0^2)$
Keeps the RCT mean close to $f(\mathbf{x})$.

Slab: $z_{h\ell} = 0$
 $\theta_{h\ell} \sim N(0, \tau_1^2)$, $\tau_1^2 > \tau_0^2$
Allows a local departure from $f(\mathbf{x})$.

$z_{h\ell} \sim \text{Bernoulli}(w)$. Leaf indicators vary. The parameters w, τ_0^2, τ_1^2 are shared.



Each leaf can stay close or depart

A leaf contribution to $g = f_1 - f$ follows

$$\theta_{hel} \mid w, \tau_0^2, \tau_1^2 \stackrel{\text{iid}}{\sim} w \text{N}(0, \tau_0^2) + (1 - w) \text{N}(0, \tau_1^2), \quad \tau_0^2 < \tau_1^2.$$

Spike Small variance τ_0^2 draws the trial-control mean toward $f(\mathbf{x})$.

Slab Large variance τ_1^2 permits a local departure from $f(\mathbf{x})$.

The indicator $z_{h\ell}$ chooses the component

$$z_{h\ell} \mid w \sim \text{Bernoulli}(w), \quad \theta_{h\ell} \mid z_{h\ell} \sim \text{N}(0, \tau_{1-z_{h\ell}}^2).$$

$z_{h\ell} = 1$ Spike variance τ_0^2 : local commensurability

$z_{h\ell} = 0$ Slab variance τ_1^2 : local conflict

In the simulations, f uses $H_f = 50$ trees and g uses $H_g = 5$ shallow trees. The application uses $H_f = 10$ and $H_g = 5$.

The prior for $f_1(\mathbf{x})$ remains centered at $f(\mathbf{x})$

At a fixed profile, each discrepancy tree contributes one leaf. If m of the H_g reached leaves select the spike,

$$f_1(\mathbf{x}) \mid f, T_g, w, \tau_0^2, \tau_1^2 \sim \sum_{m=0}^{H_g} \binom{H_g}{m} w^m (1-w)^{H_g-m} \\ \times \mathcal{N}(f(\mathbf{x}), m\tau_0^2 + (H_g - m)\tau_1^2).$$

- ▶ All components have mean $f(\mathbf{x})$.
- ▶ More slab leaves produce a wider prior for the latent trial-control mean.
- ▶ Observation variance remains separate.

Posterior computation follows standard BART moves

1. Update the trees for the shared response surface f .
 2. Update the shallow discrepancy trees for g .
 3. Draw each leaf indicator z , then its leaf mean θ .
 4. Update the spike scale τ_0^2 and mixture weight w .
- ▶ The finite Gaussian mixture gives exact collapsed factors for tree moves.
 - ▶ Log-normal AFT augmentation handles right-censored outcomes.

Exact posterior invariance does not establish finite-chain mixing.

Calibrating the amount of borrowing

Calibration targets the standardized control mean

For trial profiles $\mathbf{x}_1, \dots, \mathbf{x}_N$,

$$\mu = \frac{1}{N} \sum_{i=1}^N f_1(\mathbf{x}_i).$$

Let V_{μ}^f denote the RWD-only posterior variance of the standardized mean of f . The working prior ESS is

$$\text{ESS}_0(s_0^2) = \hat{\sigma}_1^2 \mathbb{E}_{\tau_0^2} \left[\frac{1}{V_{\mu}^f + c_g H_g \tau_0^2} \right].$$

This variance-ratio summary is motivated by ELIR (Neuenschwander et al., 2020), but it is not exact marginal ELIR for the BART scale mixture.

The fitted ceiling limits the ESS target

The largest working ESS is

$$\frac{\hat{\sigma}_1^2}{V_\mu^f}.$$

- ▶ The ceiling depends on the external-data fit at the trial profiles.
- ▶ The Pr-rule selects a scale for which working ESS is at or below the target in at least 95% of blocks.
- ▶ A target at or above the whole-chain ceiling triggers the smallest-scale override.

Equality to a scaled external sample size requires restrictive special cases.

Realized ESS uses the learned leaves

For target-profile leaf proportions $a_{h\ell}$,

$$V_g = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell}\tau_0^2 + (1 - z_{h\ell})\tau_1^2\}.$$

The realized diagnostic averages

$$\frac{\sigma_1^2}{V_\mu^f + V_g}$$

over posterior draws.

- ▶ Standardization can span several leaves in each tree.
- ▶ Regional ESS values use different standardized means and are not additive.



The borrowing map reports local compatibility

$$\mathcal{B}_c(\mathbf{x}) = P\{|g(\mathbf{x})| < c \mid \text{data}\}.$$

- ▶ c is fixed in advance on the outcome scale.
- ▶ We report the map with the posterior mean and interval for $g(\mathbf{x})$.
- ▶ In the simulations, $c = 0.5$, one third of the RCT residual SD for Gaussian outcomes.

The map is a posterior probability. It is not the number of external patients used at a profile.



The theoretical results are conditional

- Influence** The spike-and-slab leaf estimate has bounded local influence, and the spike effect vanishes under growing conflict.
- Detection** With fixed partitions, fixed f , a representable discrepancy and growing local counts, the map separates compatible and incompatible regions.
- Calibration** Under the untruncated scale law, working ESS decreases continuously from its ceiling to zero.

These results do not establish full-posterior map consistency over learned partitions. Truncation gives a positive lower ESS limit.

Simulation evidence

Simulation settings

Replicates $R = 500$ paired trials

Covariates $p = 10$ with a nonlinear outcome surface

External controls 300 patients

Gaussian trial 200 treated and 100 controls, with $\sigma_1 = 1.5$ and $\sigma_2 = 1.8$

Survival trial 225 treated and 75 controls, using log-normal AFT with about one quarter censored

ESS targets 50, 75 and 100, plus 0.25, 0.5 and 0.9 of the fitted ceiling

Five compatibility scenarios

Sc1 Source at random

Sc2 Source depends on measured covariates (X_5, X_6)

Sc3 Source and outcome depend on unmeasured U , observed through proxies

Sc4 The external-control surface shifts outside $\mathcal{R} = \{X_5 > 2\}$

Sc5 The external-control surface shifts everywhere

- ▶ Gaussian Sc4 uses $\delta_2 \in \{0.5, 1, 2\}$. Sc5 uses $\delta_2 \in \{1, 2\}$.
- ▶ Survival Sc4 uses $\delta_2 \in \{0.5, 1\}$. Sc5 uses $\delta_2 = 1$.

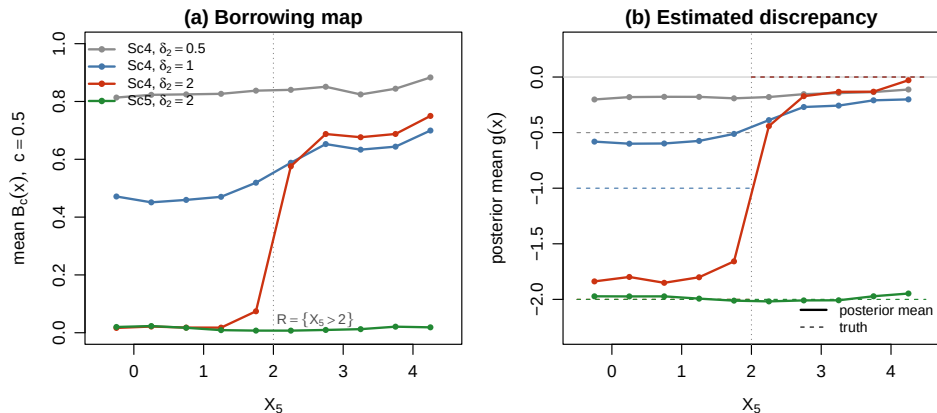
Gaussian outcomes

Scenario	LRC-BART (100)		BART-PP	
	RMSE	Coverage	RMSE	Coverage
Sc1: agreement	0.17	0.94	0.21	0.94
Sc3: unmeasured conflict	0.28	0.94	0.27	0.94
Sc4: regional, $\delta_2 = 1$	0.22	0.89	0.21	0.95
Sc4: regional, $\delta_2 = 2$	0.20	0.92	0.21	0.96
Sc5: global, $\delta_2 = 2$	0.20	0.92	0.21	0.95

- ▶ Under agreement, LRC-BART lowers RMSE by about 16% relative to BART-PP.
- ▶ Under global conflict, it suppresses borrowing and keeps bias near zero.
- ▶ A moderate regional shift remains the difficult case.

BART-PP pools trial and external controls and includes a source indicator as a predictor.

Borrowing under a regional shift



At $\delta_2 = 2$, mean compatibility is 0.02 outside $\mathcal{R}$ and 0.70 inside. The posterior mean discrepancy is -1.8 and -0.15 , compared with truths -2 and 0 .

LRC-BART target 100, first 40 replicates. The transition spans about one unit of X_5 around the boundary.

Regional conflict remains the difficult case

Moderate shift: $\delta_2 = 1$

ATE bias is -0.04 , coverage is 0.89 , and RMSE is 0.223 versus 0.207 for BART-PP.

Strong shift: $\delta_2 = 2$

ATE bias is approximately zero. Compatible-region surface RMSE is 0.824 versus 0.785 for BART-PP.

- ▶ Realized regional ESS is 9.5 inside the compatible region and 0.44 outside.
- ▶ The map is informative even when local surface estimation remains imperfect.

Surface and regional ESS comparisons use $R = 200$ paired replicates.

Survival outcomes

RMST ratio at $t^* = 3$, based on $R = 500$ trials

Scenario	LRC-BART RMSE	BART-PP RMSE
Sc1: agreement	0.097	0.106
Sc2: measured confounding	0.100	0.097
Sc3: unmeasured confounding	0.146	0.137
Sc4: regional shift	0.108	0.113
Sc5: global shift	0.112	0.110

- ▶ Under agreement, power is 0.28 for LRC-BART and 0.17 for BART-PP.
- ▶ Coverage ranges from 0.90 to 0.94 for LRC-BART across these scenarios.
- ▶ Median-ratio estimation is less favorable under unmeasured confounding.



A single-arm study cannot learn the control discrepancy

Without an internal control arm, the trial contributes no data about $g(\mathbf{x})$.

- ▶ Under no confounding, BART-CP and LRC-BART both have Gaussian RMSE 0.16.
- ▶ Under measured confounding, every method is biased: ATE bias is 1.68 for LM-CP, 0.41 for BART-CP and 0.42 for LRC-BART.
- ▶ The primary single-arm analysis fixes $w = 1$ and interprets the result as borrowing from the external-control model.

In Sc2, the fitted ceiling is 27 to 34, so an absolute target of 100 is capped.

EloKRd and the UCMM registry

Trial and registry cohorts

EloKRd phase II trial

- ▶ 30 patients, all treated with EloKRd
- ▶ 13 PFS events and 10 deaths
- ▶ 50% high-risk cytogenetics
- ▶ 30% received ASCT

UCMM registry control

- ▶ 253 patients after cohort restrictions
- ▶ Standard induction regimens
- ▶ Five baseline covariates
- ▶ ASCT recorded after induction

Estimand: standardized five-year RMST ratio

Application analysis

- ▶ Analyze PFS and OS through the standardized five-year RMST ratio.
- ▶ Fit $H_f = 10$ shared trees and $H_g = 5$ shallow discrepancy trees.
- ▶ Compare working ESS targets of 15 and 30 with a target set at 90% of the fitted ceiling.
- ▶ Report posterior intervals and $P(\text{RMST ratio} > 1)$.

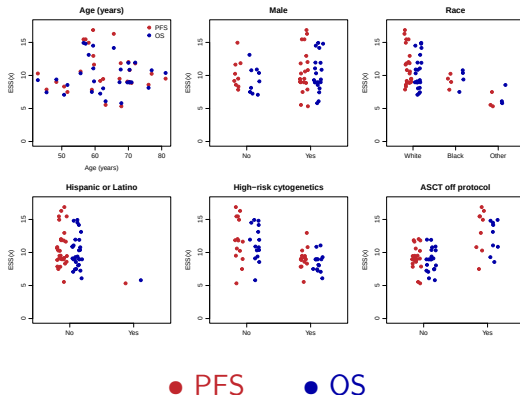
ASCT is recorded after induction. The adjusted analysis is therefore a descriptive comparison conditional on recorded profiles, not a total causal treatment effect.

EloKRd: five-year RMST ratios

Primary analysis	PFS	OS
Kaplan–Meier, unadjusted	1.01 (0.84, 1.22)	0.90 (0.79, 1.03)
Complete-pooling AFT	0.87 (0.69, 1.06)	0.80 (0.65, 0.94)
Hierarchical AFT	1.23 (0.74, 1.90)	1.21 (0.77, 1.70)
AFT-BART, no discrepancy	0.97 (0.79, 1.16)	0.90 (0.78, 1.02)
LRC-BART, $N_{\text{target}} = 30$	0.97 (0.77, 1.23)	0.90 (0.77, 1.04)
LRC-BART, $N_{\text{target}} = 15$	0.99 (0.75, 1.40)	0.91 (0.76, 1.11)
LRC-BART, $0.9 \times$ ceiling	0.97 (0.79, 1.15)	0.90 (0.77, 1.04)

At target 30, $P(\text{ratio} > 1)$ is 0.37 for PFS and 0.08 for OS. Both nonparametric intervals include one.

Local ESS at the trial profiles



Each dot is the local $ESS(x)$ for one of the 30 EloKRd profiles. Higher values mean that more UCMM control information is transferred for that patient profile.

The same 30 profiles appear in every panel. Horizontal jitter for categorical covariates only separates overlapping dots. Values range from 5 to 17 for PFS and from 6 to 15 for OS; local values are not additive.

Main findings

1. A sparse discrepancy ensemble lets borrowing vary across patient profiles.
2. The ESS calibration targets the standardized control mean and respects a fitted ceiling.
3. Simulations show a precision gain under agreement and strong protection under global conflict. Moderate regional shifts remain the hard case.

The map shows where the model regards the sources as compatible.



Limitations

- ▶ Map consistency is conditional on fixed partitions and a fixed shared surface.
- ▶ The ESS calibration is a working variance summary rather than exact marginal ELIR.
- ▶ A single-arm trial cannot identify the control discrepancy.
- ▶ Confirmatory use requires pre-specified operating-characteristic evaluation and a defined role for the borrowing map.

Multiple external sources also require joint calibration. Additive ESS ceilings have not been established.

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Technical backup

Leaf hyperpriors and nested cases

$$w \sim \text{Beta}(1, 1), \quad \tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2) \text{ truncated to } (0, \tau_1^2),$$

with τ_1^2 fixed by the range rule.

$w = 1$ Commensurate BART (C-BART), with one learned spike scale

$w = 1, \tau_0^2 \rightarrow 0$ Complete pooling BART (BART-CP)

$w = 0$ A free source offset, analogous to BART with a source indicator (BART-PP)

Appendix G.6 compares constrained joint prior modes over integer slab counts. It does not establish posterior indicator probabilities, which also depend on the likelihood and partitions.



Why use separate ensembles?

- ▶ The shared surface f and discrepancy g can use different partition and shrinkage priors.
- ▶ The simulations use $H_f = 50$ trees and $H_g = 5$ shallow trees, with split prior $(0.5, 3)$ for g .
- ▶ At fixed leaf variances and partitions, fewer discrepancy trees reduce the all-spike variance at a profile.
- ▶ The slab range rule also depends on H_g .

The simulation results assess this specification. They do not show that a shared-leaf model cannot detect regional discrepancies.

Calibration conventions and borrowing summaries

- ▶ The constant c_g averages squared leaf proportions before inversion. The calibration scale law is untruncated.
- ▶ The working ESS and ceiling use the standardized control mean. The pointwise ESS uses a profile-specific posterior variance.
- ▶ The realized regional ESS uses posterior indicators and leaf proportions for its own standardized regional mean.
- ▶ $B(x)$, the probability that all reached leaves select the spike, remains a diagnostic. Its prior scale is w^{H_g} .
- ▶ A fixed positive spike scale need not identify compatibility through $B(x)$ as local RCT counts grow.

Compatibility probability, prior ESS and realized regional ESS answer different questions. None is an observed count of external patients used.



Comparator labels

- LM-PP** Linear regression on pooled controls, with source indicator
- BART-NP** Standard BART fitted to the RCT controls only
- BART-CP** Standard BART fitted to pooled controls, without a source indicator
- BART-PP** Standard BART fitted to pooled controls, with a source indicator
- C-BART** LRC-BART with $w = 1$, using one commensurate discrepancy scale

The study also includes linear/AFT analogues, PSCL, MAP and hierarchical models.



- ▶ Under Sc1, LRC-BART RMSE is 0.172 versus 0.205 for BART-PP. Prior ESS is 82, the ceiling is 127, and realized ESS is 53.
- ▶ Under Sc3, realized ESS falls to 1.6. Bias exceeds BART-PP by 0.012 with paired SE 0.001.
- ▶ Under Sc5 at $\delta_2 = 2$, the map averages 0.01 and realized ESS is 0.2.
- ▶ Under Sc2, the ceiling falls from 127 to 23, so all absolute targets are capped.
- ▶ Under Sc4 at $\delta_2 = 2$, surface RMSE away from the boundary is 0.747 versus 0.723, a difference of 0.024 with SE 0.005 over $R = 200$ replicates.

Additional borrowing-map results

- ▶ At $\delta_2 = 1$, the mean map is 0.62 inside $\mathcal{R}$ and 0.52 outside. The posterior mean discrepancy is -0.25 and -0.60 .
- ▶ At $\delta_2 = 0.5$, the map is nearly flat at 0.84, below the stated resolution.
- ▶ Under the global shift Sc5 at $\delta_2 = 2$, the map is 0.01 throughout the covariate space.

A secondary Sc4 shift on the null covariate X_7 gives similar map and ATE results.

Exact leaf marginals for tree moves

Conditionally on (τ_0^2, τ_1^2, w) ,

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp \left\{ \frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)} \right\},$$

$$M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)}.$$

Here A and B are precision-weighted counts and sums from both sources in an f leaf. The terms $\bar{y}_1$ and $v_1 = \sigma_1^2/n_1$ are the RCT residual mean and variance in a g leaf.

Grow, prune and change moves use these collapsed ratios. One sweep costs a standard BART fit on $n_1 + n_2$ observations plus a five-tree ensemble on n_1 .

Conditional bounds and consistency statements

Write $v = \sigma_1^2/n$ for an RCT residual-mean variance. The results condition on f , other discrepancy trees, partitions, scales and $0 < w < 1$.

Let $a = \log\{(1-w)/w\} + \frac{1}{2} \log\{(\tau_0^2 + v)/(\tau_1^2 + v)\}$ and $C = e^{-a}$.

Influence Relative to the slab estimate, $\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}$. The shift vanishes at a Gaussian rate under growing conflict.

Map With fixed partitions, fixed f , a representable discrepancy and growing local counts, $\mathcal{B}_c(x) \rightarrow \mathbb{I}\{|d| < c\}$ for $|d| \neq c$.

ESS Under the untruncated law, ESS_0 is continuous, convex and strictly decreasing from its ceiling to zero. Truncation gives a positive lower limit.

A fixed positive spike scale need not identify compatibility through the all-spike indicator. Grid-selection consistency requires the stated moment and separation conditions.

Gaussian outcomes: full comparison

Method	Sc1			Sc3, $\rho = 0$			Sc4, $\delta_2 = 1$			Sc4, $\delta_2 = 2$			Sc5, $\delta_2 = 2$		
	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov
LM-PP	-0.00	0.34	0.94	0.04	0.35	0.95	-0.01	0.34	0.95	-0.02	0.34	0.95	-0.04	0.34	0.94
BART-NP	0.00	0.22	0.91	-0.00	0.29	0.94	0.00	0.22	0.91	0.00	0.22	0.91	0.00	0.22	0.91
BART-CP	0.01	0.14	0.96	0.98	1.00	0.00	-0.34	0.37	0.35	-0.68	0.70	0.00	-1.39	1.40	0.00
BART-PP	0.01	0.21	0.94	0.03	0.27	0.94	-0.00	0.21	0.95	-0.01	0.21	0.96	-0.01	0.21	0.95
C-BART	0.01	0.15	0.95	0.23	0.38	0.87	-0.21	0.27	0.73	-0.23	0.39	0.66	-0.32	0.65	0.76
LRC-BART (100)	0.01	0.17	0.94	0.04	0.28	0.94	-0.04	0.22	0.89	0.00	0.20	0.92	-0.00	0.20	0.92

CAHB comparisons use the first $R = 200$ paired Gaussian replicates and assess transfer of kernel-local estimation. They do not establish a general method ranking (Jin et al., 2023).

Survival outcomes: full RMST results

RMST ratio at $t^* = 3$, based on $R = 500$ trials

Scenario	LRC-BART (100)			BART-PP		
	Bias	RMSE	Cov	Bias	RMSE	Cov
Sc1: agreement	0.003	0.097	0.94	-0.023	0.106	0.94
Sc2: measured confounding	0.018	0.100	0.93	-0.002	0.097	0.95
Sc3: unmeasured confounding	0.031	0.146	0.94	-0.005	0.137	0.92
Sc4: regional shift	-0.037	0.108	0.90	-0.059	0.113	0.91
Sc5: global shift	0.001	0.112	0.93	-0.041	0.110	0.93

Sc3 uses $\rho = 0$. Sc4 and Sc5 use $\delta_2 = 1$. Full results appear in the paper and Web Table S2.

Single-arm sensitivity details

The trial contributes treated subjects only ($n_T = 200$ Gaussian and 30 or 200 survival), and the 300 external controls are the sole control source.

- ▶ Under no confounding, the survival RMST ratio at $n_T = 30$ has RMSE 0.16 to 0.19 and coverage 0.92 to 0.97.
- ▶ The fractional ESS target widens intervals. Coverage ranges from 0.82 to 0.99.
- ▶ With $w < 1$, slab variance adds uncertainty: SD 1.4 for the ATE and 10 to 34 for the median ratio.
- ▶ A zero-mean discrepancy preserves linear log-time means, but survival-ratio means can change.
- ▶ Hierarchical regressions give ATE SD 8 under the stated hyperprior.